

Negative Exponents with Variable Bases

MA.8.AR.1.1

I can rewrite an algebraic expression with negative exponents using only positive exponents, and apply several laws of exponents to a monomial expression.

Yesterday every exponent on a letter was positive.

Today x^3 arrives — and it is not a negative number.

Warm-Up

On your own. Four minutes. No notes.

1. Write as a single monomial: $(3x^2)^3$
2. Find the value: 5^{-2}
3. Which is greater, $\sqrt{80}$ or 9?
4. What is the reciprocal of $\frac{2}{3}$?

Warm-Up

Answers.

1. Write as a single monomial: $(3x^2)^3$ **$27x^6$**
2. Find the value: 5^{-2} **$1/25$**
3. Which is greater, $\sqrt{80}$ or 9? **9**
4. What is the reciprocal of $\frac{2}{3}$? **$\frac{3}{2}$**

Notes

Copy the table.

I. The pattern, with a letter

$\mathbf{x^4}$	$\mathbf{x^3}$	$\mathbf{x^2}$	$\mathbf{x^1}$	$\mathbf{x^0}$	$\mathbf{x^{-1}}$	$\mathbf{x^{-2}}$	$\mathbf{x^{-3}}$
$\mathbf{x \cdot x \cdot x \cdot x}$	$\mathbf{x \cdot x \cdot x}$	$\mathbf{x \cdot x}$	$\mathbf{x}$	$\mathbf{1}$	$\mathbf{1/x}$	$\mathbf{1/x^2}$	$\mathbf{1/x^3}$

$$x^{-n} = \frac{1}{x^n} \qquad \frac{1}{x^{-n}} = x^n \qquad (x \neq 0)$$

Notes

II. Moving factors across the fraction bar

$$3x^{-2} = \frac{3}{x^2}$$

$$y^{-3}z^{-4} = \frac{1}{y^3z^4}$$

$$\frac{a^2}{b^{-3}} = a^2b^3$$

$$\frac{5x^{-4}y^2}{z^{-1}} = \frac{5y^2z}{x^4}$$

Only the factor with the negative exponent moves. Its exponent turns positive.

Notes

III. The laws with negative exponents

$$x^4 \cdot x^{-6} = x^{4+(-6)} = x^{-2} = \frac{1}{x^2} \qquad \frac{m^3}{m^7} = m^{3-7} = m^{-4} = \frac{1}{m^4}$$

$$(2y^{-3})^2 = 2^2 \cdot y^{-6} = \frac{4}{y^6}$$

Apply the law. Then rewrite so that every exponent is positive.

Example 1

Multiplying monomials with negative exponents.

$$(2x^{-3}y^2)(6x^5y^{-4})$$

Write an equivalent expression with positive exponents only.

Example 1

Worked.

$$(2x^{-3}y^2)(6x^5y^{-4}) = 12 \cdot x^{-3+5} \cdot y^{2+(-4)} \quad \begin{array}{l} \text{multiply coefficients;} \\ \text{add exponents} \end{array}$$

$$= 12x^2y^{-2} = \frac{12x^2}{y^2} \quad \text{move } y^{-2} \text{ down; the 12 and the } x^2 \text{ stay up}$$

Answer: $12x^2/y^2$

Your Turn

Same steps, your numbers. Boards up when done.

$$(4a^{-2}b)(a^5b^{-3})$$

Your Turn

Answer.

$$(4a^{-2}b)(a^5b^{-3})$$

$$4 \cdot a^3 \cdot b^{-2}$$

$$\textbf{Answer: } \frac{4a^3}{b^2}$$

Example 2

Several laws in one expression — the state guide's item.

$$\left(\frac{a^3}{4b^{-7}}\right)^5$$

Write an equivalent expression with positive exponents only.

Example 2

Worked.

$$\left(\frac{a^3}{4b^{-7}}\right)^5 = \frac{a^{15}}{4^5 b^{-35}} \quad \text{raise every factor: } a^3, 4, b^{-7}$$
$$= \frac{a^{15} b^{35}}{1,024} \quad 4^5 = 1,024; b^{-35} \text{ moves up as } b^{35}$$

Answer: $a^{15}b^{35}/1,024$

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(\frac{x^2}{3y^{-2}}\right)^3$$

Your Turn

Answer.

$$\left(\frac{x^2}{3y^{-2}} \right)^3$$

x^6 over $27y^6$, then move the y up

Answer: $\frac{x^6 y^6}{27}$

Whiteboards • Question 1 of 9

Boards up on three.

$$x^{-4}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 1 of 9

Answer.

$$x^{-4}$$

reciprocal of x^4

Answer: $\frac{1}{x^4}$

Whiteboards • Question 2 of 9

Boards up on three.

$$\frac{1}{y^{-5}}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 2 of 9

Answer.

$$\frac{1}{y^{-5}}$$

the power moves up

Answer: y^5

Whiteboards • Question 3 of 9

Boards up on three.

The brightness of a lamp from x meters away is $3x^{-2}$ units.

The lamp is 15 centimeters tall.

How bright is the lamp from 2 meters away?

Answer it.

Whiteboards • Question 3 of 9

Answer.

The brightness of a lamp from x meters away is $3x^{-2}$ units.

The lamp is 15 centimeters tall.

How bright is the lamp from 2 meters away?

$$3 \cdot 2^{-2} = 3 \text{ over } 2^2$$

Answer: $\frac{3}{4}$

Whiteboards • Question 4 of 9

Boards up on three.

$$y^{-3}z^{-4}$$

Answer it.

Rewrite with positive exponents.

Whiteboards • Question 4 of 9

Answer.

$$y^{-3}z^{-4}$$

both factors move down

$$\text{Answer: } \frac{1}{y^3 z^4}$$

Whiteboards • Question 5 of 9

Boards up on three.

$$x^4 \cdot x^{-6}$$

Which expression is equivalent?

A. $-x^2$

B. $1/x^{24}$

C. $1/x^2$

D. x^{10}

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$x^4 \cdot x^{-6}$$

Which expression is equivalent?

A. $-x^2$

B. $1/x^{24}$

C. $1/x^2$

D. x^{10}

Answer: C — $1/x^2$

Whiteboards • Question 6 of 9

Boards up on three.

$$(2x^{-3})^2$$

Which expression is equivalent?

A. $4/x$

B. $1/(4x^6)$

C. $2/x^6$

D. $4/x^6$

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$(2x^{-3})^2$$

Which expression is equivalent?

A. $4/x$

B. $1/(4x^6)$

C. $2/x^6$

D. $4/x^6$

Answer: D — $4/x^6$

Whiteboards • Question 7 of 9

Boards up on three.

$$\frac{m^3}{m^7}$$

Answer it.

Write with a positive exponent.

Whiteboards • Question 7 of 9

Answer.

$$\frac{m^3}{m^7}$$

m^{-4} — then rewrite

Answer: $\frac{1}{m^4}$

Whiteboards • Question 8 of 9

Boards up on three.

Find the value of n :

$$x^{-5} \cdot x^n = x^2$$

Answer it.

Which law applies?

Whiteboards • Question 8 of 9

Answer.

Find the value of n :

$$x^{-5} \cdot x^n = x^2$$

$$-5 + n = 2$$

Answer: $n = 7$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

How are x^3 and x^{-3} related?

Explain, and give the value of each when $x = 2$.

Write your answer in sentences.

Use the word reciprocal.

Whiteboards · Question 9 of 9

Answer.

How are x^3 and x^{-3} related?

Explain, and give the value of each when $x = 2$.

reciprocals: 8 and $1/8$

Answer: x^{-3} is the reciprocal of x^3 . When $x = 2$: $x^3 = 8$ and $x^{-3} = 1/8$.

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Variables with negative exponents — 2NL

Multiply and divide powers: variable bases — 85P